AI Engineering
Complete 12-Week Syllabus

From LLM foundations to production-grade agentic systems

Day 1: Monday 12 October 2026
Day 84: Sunday 3 January 2027
84 consecutive teaching days · no rest days

Current to October 2026 — 2026 Currency Layer (u1) verified 11 Oct 2026
Merged syllabus m1 — 133/133 topics and 53/53 resources preserved verbatim

Instructor: Pro AI Principal Architect

github.com/irudi2021/ai-engineering-journey

SCHEDULE — LOCKED

> **This file is the single source of truth for all dates.** If any other document disagrees with this file, this file wins. Content authority: CURRICULUM_MASTER.md.

Status: LOCKED — 11 October 2026 · FRESH RUN

The Run in One Line

Day 1: Monday 12 October 2026 → Day 84: Sunday 3 January 2027 — 84 consecutive teaching days, no rest days.

Key Dates

Milestone	Date	Day
Start	**Monday, 12 October 2026**	**Day 1**
End of Week 4	Sunday, 8 November 2026	Day 28
Christmas — teaching day	Friday, 25 December 2026	Day 75
New Year — teaching day	Friday, 1 January 2027	Day 82
End	**Sunday, 3 January 2027**	**Day 84**

Every week runs Monday–Sunday. No pauses, no rest days — Christmas and New Year are teaching days by choice.

Week Overview

Week	Days	Dates	Theme	Project
01	1–7	Oct 12–18	Foundations — LLM internals	Model Selection Dashboard (MINI)
02	8–14	Oct 19–25	Efficiency — quantization, fine-tuning, serving	Domain Fine-Tune + Quantize (PORTFOLIO)
03	15–21	Oct 26–Nov 1	Retrieval & Agents — RAG, agents, multi-agent	End-to-End Text-to-SQL (CAPSTONE)
04	22–28	Nov 2–8	Production — LLMOps, versioning, cost, security, multimodal	Production Hardening & Review (PORTFOLIO)
05	29–35	Nov 9–15	Agents, Harness & Loop Engineering + Agentic Frameworks	AI Research Agent with Tool Calling (PORTFOLIO)
06	36–42	Nov 16–22	Context Engineering, MCP & Multi-Agent Frameworks	Multi-Agent Business Analyst System (PORTFOLIO)
07	43–49	Nov 23–29	Production AI Systems, Evaluation & Observability	Eval & Observability Platform (PORTFOLIO)
08	50–56	Nov 30–Dec 6	AI Application Engineering (FastAPI) & Agentic System Design	End-to-End AI Backend + Agent Runtime (PORTFOLIO)
09	57–63	Dec 7–13	Machine Learning & Deep Learning Foundations	ML & DL Project Development — 5 projects (PORTFOLIO)
10	64–70	Dec 14–20	NLP, Multimodal, Reasoning & Red Teaming	NLP & Multimodal Projects + Red Team — 3 projects (PORTFOLIO)
11	71–77	Dec 21–27	Cloud Deployment, AI Infrastructure & LLMOps	Deploy a Production-Ready AI Service (PORTFOLIO)
12	78–84	Dec 28–Jan 3	No-Code AI Platforms & Enterprise Capstone	Enterprise Multi-Document RAG Assistant (CAPSTONE)

Track 0 (self-study, not teaching days): Python, Pandas, NumPy, OOP & SQL — preserved in full in `CURRICULUM_MASTER.md`.

Day-by-Day

Each day: the core topic plus its **2026 addition** (Currency Layer u1, verified 11 Oct 2026).

Week 01 · Foundations — LLM internals — Days 1–7 · Mon 12 Oct – Sun 18 Oct 2026

Day	Date	Topic	2026 addition
1	Mon Oct 12	Tokenization	Tokenizer differences across model families; token-cost tax on non-English text (Tamil vs English)
2	Tue Oct 13	Embeddings	Matryoshka (truncatable) embeddings; multi-vector late interaction (ColBERT); multimodal embeddings
3	Wed Oct 14	Transformer Architecture	Mixture-of-Experts (total vs active parameters); GQA / MLA attention; long-context mechanics
4	Thu Oct 15	Training Lifecycle	The 2026 post-training stack: SFT → preference optimisation (DPO) → RL with verifiable rewards (RLVR / GRPO); how reasoning models are made
5	Fri Oct 16	Scaling Laws & Prompt Engineering	Test-time compute scaling; reasoning-effort controls; prompting reasoning models vs classic models
6	Sat Oct 17	Structured Output	Constrained decoding and strict JSON-Schema output modes; JSON Schema 2020-12
7	Sun Oct 18	**Week 1 Project** — Model Selection Dashboard	Live 2026 model landscape (frontier vs open-weight, verified from official changelogs on the day); model churn and migration cost; model routing

Week 02 · Efficiency — quantization, fine-tuning, serving — Days 8–14 · Mon 19 Oct – Sun 25 Oct 2026

Day	Date	Topic	2026 addition
8	Mon Oct 19	Quantization	FP8 and 4-bit floating-point formats (NVFP4 / MXFP4); KV-cache quantisation
9	Tue Oct 20	LoRA & PEFT	QLoRA; serving many LoRA adapters on one base model (multi-LoRA)
10	Wed Oct 21	Full Fine-Tuning	RL fine-tuning: RLVR / GRPO, verifier and reward design; open tooling (TRL, verl, OpenRLHF)
11	Thu Oct 22	Model Serving	vLLM, SGLang, TensorRT-LLM; prefill/decode disaggregation; llm-d (CNCF Sandbox, Mar 2026) and NVIDIA Dynamo; speculative decoding; prefix caching
12	Fri Oct 23	Evaluation	Benchmark saturation and contamination; agentic benchmarks (e.g. SWE-bench Verified); building your own task evals
13	Sat Oct 24	Efficiency	Prompt caching, batch APIs, KV-cache reuse, model cascades
14	Sun Oct 25	**Week 2 Project** — Text-to-SQL Foundation (running capstone begins)	Project add-on: eval set + cost report per candidate model

Week 03 · Retrieval & Agents — RAG, agents, multi-agent — Days 15–21 · Mon 26 Oct – Sun 1 Nov 2026

Day	Date	Topic	2026 addition
15	Mon Oct 26	RAG Foundations	Decision framework: RAG vs long context vs agentic search
16	Tue Oct 27	RAG Indexing Pipelines	Contextual chunk enrichment; late chunking; layout-aware document parsing
17	Wed Oct 28	RAG Retrieval Deep Dive	Multi-vector retrieval; GraphRAG; query rewriting and routing
18	Thu Oct 29	RAG Generation & Prompt Engineering	Grounded citations; agentic RAG (the agent decides when and what to retrieve)
19	Fri Oct 30	Agents and Tool Use	Tool calling with reasoning models; computer-use and browser agents
20	Sat Oct 31	Multi-Agent Systems	Orchestrator-worker and sub-agent patterns; preview of the A2A protocol
21	Sun Nov 1	**Week 3 Capstone** — End-to-End Text-to-SQL	Capstone add-on: tool-call trace review and failure taxonomy

Week 04 · Production — LLMOps, versioning, cost, security, multimodal — Days 22–28 · Mon 2 Nov – Sun 8 Nov 2026

Day	Date	Topic	2026 addition
22	Mon Nov 2	Production LLM Operations	OpenTelemetry GenAI semantic conventions (still 'Development' status in 2026); AI gateways
23	Tue Nov 3	Prompt and Model Versioning	Prompt registries; model-deprecation and migration playbook for fast model churn
24	Wed Nov 4	Cost Optimization at Scale	Prompt caching, batch inference, routing and reasoning-effort budgets as cost levers
25	Thu Nov 5	Security and Compliance	OWASP Top 10 for LLM Applications (2025) + OWASP Top 10 for Agentic Applications 2026 (ASI01-ASI10, published 9 Dec 2025); indirect prompt injection
26	Fri Nov 6	Multi-Modal Systems	Real-time voice agents (speech-to-speech); video understanding
27	Sat Nov 7	Fine-Tuning for Production	Distillation from frontier models; synthetic data; fine-tune vs context-engineer decision
28	Sun Nov 8	**Week 4 Capstone** — Production Hardening & Review	Review add-on: OWASP LLM + Agentic risk mapping of the capstone

Week 05 · Agents, Harness & Loop Engineering + Agentic Frameworks — Days 29–35 · Mon 9 Nov – Sun 15 Nov 2026

Day	Date	Topic	2026 addition
29	Mon Nov 9	Agent Architectures Beyond ReAct	'Agent = Model + Harness' (Bockeler, Thoughtworks, Apr 2026); long-horizon / deep agents; planning patterns
30	Tue Nov 10	**Harness Engineering I — Tool Design**	Agent Skills open standard (SKILL.md, progressive disclosure, agentskills.io); AGENTS.md project instructions
31	Wed Nov 11	**Harness Engineering II — Responses, Errors & Environment**	Sandboxing and containment for agents with shell, web and file access; least-privilege tool permissions
32	Thu Nov 12	**Loop Engineering I — Termination & Budgets**	Token, time and cost budgets per run; reasoning-effort as a loop control
33	Fri Nov 13	**Loop Engineering II — Recovery, Sub-Agents & Human-in-the-Loop Design**	Approval gates and permission modes; interrupt / resume for human review
34	Sat Nov 14	**Agentic Frameworks I — LangChain Fundamentals & LangGraph Workflows**	LangGraph durable execution, checkpointing and interrupts
35	Sun Nov 15	**Week 5 Project** — AI Research Agent with Tool Calling	Project add-on: trajectory-level eval of the research agent

Week 06 · Context Engineering, MCP & Multi-Agent Frameworks — Days 36–42 · Mon 16 Nov – Sun 22 Nov 2026

Day	Date	Topic	2026 addition
36	Mon Nov 16	**Context Engineering I — The Context Budget & Token Management**	Context engineering framework: Write / Select / Compress / Isolate
37	Tue Nov 17	**Context Engineering II — Compaction & Progressive Disclosure**	Compaction, memory files and skills as progressive disclosure in practice
38	Wed Nov 18	MCP (Model Context Protocol) Fundamentals	MCP spec revision 2026-07-28: stateless core (no initialize handshake, no sessions), server/discover, Multi Round-Trip Requests, Tasks extension, Roots/Sampling/Logging deprecated; MCP governed by the Agentic AI Foundation (Linux Foundation, Dec 2025)
39	Thu Nov 19	Building a Production MCP Server	MCP authorisation (OAuth, Client ID Metadata Documents), cacheable list results (ttlMs / cacheScope), OpenTelemetry trace propagation, migrating a 2025-11-25 server
40	Fri Nov 20	**Agentic Frameworks II — CrewAI, AG2 / AutoGen / Agno & Multi-Agent Collaboration**	A2A protocol v1.0 (Linux Foundation; Agent Cards, task lifecycle, JSON-RPC + gRPC); MCP vs A2A; vendor agent SDKs (OpenAI Agents SDK, Claude Agent SDK, Google ADK, Microsoft Agent Framework)
41	Sat Nov 21	Agent Memory Architectures & Planning Strategies	Long-term memory stores; memory as a tool; memory poisoning risk (OWASP Agentic)
42	Sun Nov 22	**Week 6 Project** — Multi-Agent Business Analyst System	Project add-on: expose one agent over A2A and its tools over MCP

Week 07 · Production AI Systems, Evaluation & Observability — Days 43–49 · Mon 23 Nov – Sun 29 Nov 2026

Day	Date	Topic	2026 addition
43	Mon Nov 23	**AI System Architecture & Orchestration**	Reference architecture: gateway → guardrails → orchestrator → tools → observability
44	Tue Nov 24	LLM-as-a-Judge & Eval Harness Design	Eval-driven development; golden datasets; agent trajectory and tool-use evals
45	Wed Nov 25	**Hallucination Detection & Evaluation**	Groundedness and citation checks; uncertainty signals
46	Thu Nov 26	**RAG Evaluation — RAGAS & TruLens**	Agentic RAG evaluation (retrieval decisions, not just answers)
47	Fri Nov 27	**LangSmith for Debugging & Observability**	Vendor-neutral tracing with OpenTelemetry GenAI conventions alongside LangSmith (Langfuse, Arize Phoenix)
48	Sat Nov 28	**AI Safety & Guardrails in Production**	Guardrails mapped to OWASP Agentic risks; runtime policy enforcement
49	Sun Nov 29	**Week 7 Project** — Eval & Observability Platform	Project add-on: CI eval gate that blocks a regression

Week 08 · AI Application Engineering (FastAPI) & Agentic System Design — Days 50–56 · Mon 30 Nov – Sun 6 Dec 2026

Day	Date	Topic	2026 addition
50	Mon Nov 30	**AI Application Architecture · REST APIs, HTTP Methods & API Integration · JSON, JSON Schema & Structured Data**	Streaming tool-call events; structured outputs at the API boundary
51	Tue Dec 1	**FastAPI Fundamentals & API Development · Request Handling, Response Models & Error Handling**	Async FastAPI patterns for long-running LLM calls
52	Wed Dec 2	**Authentication, API Keys & Environment Variables · File Uploads, Streaming Responses & Async APIs**	Server-Sent Events streaming; secrets management for model API keys
53	Thu Dec 3	**Database Integration (SQLite/PostgreSQL Basics) · AI API Integration (OpenAI, Gemini, Claude)**	Multi-provider abstraction and fallbacks; provider-native MCP connectors
54	Fri Dec 4	Agentic System Design — Runtime, State Management & Queueing	Durable execution for long-running agents (workflow engines such as Temporal); resumable state
55	Sat Dec 5	Scaling, Backpressure, Failure Isolation & Concurrency	Rate limits, retries with jitter, circuit breakers for model providers
56	Sun Dec 6	**Week 8 Project** — End-to-End AI Backend Development using FastAPI + Scalable Agent Runtime	Project add-on: provider failover drill

Week 09 · Machine Learning & Deep Learning Foundations — Days 57–63 · Mon 7 Dec – Sun 13 Dec 2026

Day	Date	Topic	2026 addition
57	Mon Dec 7	**ML Fundamentals — Types of ML (Supervised, Unsupervised & Reinforcement) · Features, Labels, Training/Testing Data & Train/Test Split · Classification, Regression & Clustering Algorithms**	Where classical ML still runs FAANG-scale systems (ranking, recommendations, fraud)
58	Tue Dec 8	**Model Validation using Cross-Validation · Evaluation Metrics: Accuracy, Precision, Recall, F1-Score & ROC-AUC · Confusion Matrix & Handling Class Imbalance**	Metric choice for imbalanced production data
59	Wed Dec 9	**Overfitting, Underfitting & Regularization · Feature Engineering & Data Preprocessing · Feature Scaling, Normalization & Data Preparation · End-to-End ML Workflow using Scikit-learn**	Data leakage and train/serve skew
60	Thu Dec 10	**Deep Learning Fundamentals — Neural Networks & Perceptron · Activation Functions, Loss Functions & Optimizers · Forward Propagation, Backward Propagation & Gradient Descent**	From perceptron to transformer: the same gradient-descent engine
61	Fri Dec 11	**Epochs, Batch Size, Learning Rate & Hyperparameter Tuning · Batch Normalization, Dropout & Vanishing Gradient Problem**	Mixed-precision training basics (FP16 / BF16)
62	Sat Dec 12	**CNNs & Image Classification · Pretrained Models, Transfer Learning & Fine-Tuning · SOTA Models & Modern DL Applications · End-to-End Model Development using TensorFlow/Keras or PyTorch**	Vision Transformers (ViT) and diffusion models in context
63	Sun Dec 13	**Week 9 Project** — ML & DL Project Development & Deployment	Project add-on: model card for each project

Week 10 · NLP, Multimodal, Reasoning & Red Teaming — Days 64–70 · Mon 14 Dec – Sun 20 Dec 2026

Day	Date	Topic	2026 addition
64	Mon Dec 14	**NLP Foundations — Introduction to NLP & Text Preprocessing · Tokenization, Stop Words, Stemming & Lemmatization**	Why classical preprocessing still matters before LLM pipelines (cost, PII scrubbing)
65	Tue Dec 15	**Bag of Words (BoW), Count Vectorizer & TF-IDF · N-Grams, POS Tagging & Named Entity Recognition (NER)**	Classical NER vs LLM extraction: cost and accuracy trade-off
66	Wed Dec 16	**Word Embeddings: Word2Vec, GloVe & Vector Embeddings · End-to-End NLP Pipeline using Hugging Face Transformers**	Hugging Face Hub model selection; small open models for on-device and edge
67	Thu Dec 17	Vision-Language Models, OCR Pipelines & Document Understanding at Scale	Document AI with vision-language models replacing OCR-only pipelines
68	Fri Dec 18	Chart & Diagram Reasoning · Reasoning Models & Test-Time Compute	Reasoning models in production: effort settings, latency/cost trade-offs, when not to use them
69	Sat Dec 19	Red Teaming, OWASP LLM Top 10 & Responsible AI Governance	EU AI Act after the Digital Omnibus (Reg. (EU) 2026/1744, in force 27 Jul 2026): Annex III high-risk → 2 Dec 2027, Annex I → 2 Aug 2028, Art. 50 transparency still from Aug 2026; NIST AI RMF; ISO/IEC 42001; OWASP Agentic red-team cases
70	Sun Dec 20	**Week 10 Project** — NLP & Multimodal Projects + Red Team Assessment	Project add-on: red-team agent tool misuse, not only prompt injection

Week 11 · Cloud Deployment, AI Infrastructure & LLMOps — Days 71–77 · Mon 21 Dec – Sun 27 Dec 2026

Day	Date	Topic	2026 addition
71	Mon Dec 21	**Introduction to AWS, Azure & Google Cloud Platform · Compute Services (EC2, Azure VM, Compute Engine) · Storage Services (S3, Blob Storage, Cloud Storage)**	Managed agent platforms: Amazon Bedrock AgentCore, Microsoft Azure AI Foundry, Google Vertex AI agent tooling
72	Tue Dec 22	**Docker Fundamentals & Containerization · Docker for AI Applications**	GPU containers and model-weight caching
73	Wed Dec 23	**Deploying FastAPI Applications · Deploying AI Applications on Cloud · Serverless AI Deployments**	Serverless GPU inference; cold-start trade-offs
74	Thu Dec 24	**CI/CD Basics for AI Applications · Monitoring & Logging · Cost Optimization & Cloud Best Practices**	Eval gates inside CI/CD; OpenTelemetry-based AI monitoring
75	Fri Dec 25	LLMOps — Model Registries, Automated Eval Gates & Inference Optimisation (TensorRT-LLM, ONNX)	Kubernetes-native inference (llm-d, NVIDIA Dynamo) and prefill/decode disaggregation at scale
76	Sat Dec 26	Synthetic Data Generation (Self-Instruct, Evol-Instruct) · Feature Stores & Data Lineage · Cost Engineering & Chargeback	Synthetic data for evals and RL environments; verifier quality as the ceiling
77	Sun Dec 27	**Week 11 Project** — Deploy a Production-Ready AI Service	Project add-on: cost-per-request dashboard

Week 12 · No-Code AI Platforms & Enterprise Capstone — Days 78–84 · Mon 28 Dec – Sun 3 Jan 2027

Day	Date	Topic	2026 addition
78	Mon Dec 28	**Introduction to No-Code & Low-Code AI Platforms · Claude Projects & Claude Artifacts · ChatGPT Projects, GPTs & Canvas**	Agent builders inside productivity suites; when no-code agents are enough
79	Tue Dec 29	**Microsoft Copilot & Microsoft 365 Copilot · Microsoft Copilot Studio & AI Agents (Basics) · Google Gemini Workspace & Gems**	Enterprise agent governance in Microsoft 365 / Google Workspace
80	Wed Dec 30	**AI Workflow Automation using n8n (Introduction) · AI Coding Assistants (GitHub Copilot, Cursor & Windsurf) · Rapid AI Prototyping & Business Use Cases · Choosing Between No-Code, Low-Code & Custom AI Development**	Coding agents as the main developer interface (Claude Code, OpenAI Codex, GitHub Copilot agent mode, Cursor); Agent Skills and AGENTS.md in a real repo
81	Thu Dec 31	**Capstone I** — Architecture, ADRs & C4 Diagrams	ADRs for MCP, A2A, model routing and eval-gate choices
82	Fri Jan 1	**Capstone II** — Build & Integration	Build with MCP tools, Agent Skills and OpenTelemetry tracing
83	Sat Jan 2	**Capstone III** — Evaluation, Security & Deployment	Eval gate + OWASP LLM/Agentic mapping + EU AI Act risk classification
84	Sun Jan 3	**Capstone IV** — Enterprise AI Assistant Final Defence	Final defence + State of AI Engineering 2026 → 2027 outlook

CURRICULUM MASTER — 12-Week AI Engineering Syllabus

LOCKED 11 October 2026 — FRESH RUN. Day 1: Monday 12 October 2026 → Day 84: Sunday 3 January 2027. 84 consecutive teaching days. No rest days — Christmas (Day 75) and New Year (Day 82) are teaching days by choice.

This document is the **complete content authority** for all 12 weeks. Dates are governed by SCHEDULE.md.

Syllabus composition: an external 4-month (16-week) syllabus was merged into this 12-week programme on 2 August 2026 (merge revision m1). **No weeks were added.** Every incoming topic and every listed resource is preserved verbatim below.

2026 Currency Layer (revision u1, 11 October 2026): every day now carries one 2026 addition so the course ends current to the state of AI engineering in October 2026. Additive only — no day title, topic or resource from m1 was changed or removed.

0. 2026 CURRENCY LAYER — What Changed in the Field and Where It Is Taught

Checked on 11 October 2026 against primary sources (listed below). Rules:

1. **Additive.** Each day keeps its m1 topic; the 2026 addition is taught inside it.
2. **Model-agnostic.** No frontier model version is hard-coded — names change monthly.

Day 7 builds the live model landscape from official provider changelogs on the day.

3. **Verified or excluded.** Anything reported by a single unverified source was left out.

2026 shift	Taught on
Agent = Model + Harness; harness engineering as a core skill	Days 29–33
Agent Skills open standard (SKILL.md) and AGENTS.md	Days 30, 37, 80
Context engineering: Write / Select / Compress / Isolate	Days 36–37
MCP spec revision 2026-07-28 (stateless core, MRTR, Tasks extension)	Days 38–39
MCP, goose and AGENTS.md under the Agentic AI Foundation (Linux Foundation)	Day 38
A2A protocol v1.0 and MCP-vs-A2A	Days 20, 40, 42
OWASP Top 10 for Agentic Applications 2026 (ASI01–ASI10)	Days 25, 41, 48, 69, 83
RL with verifiable rewards (RLVR / GRPO) and reasoning models	Days 4, 5, 10, 68
Prefill/decode disaggregation, llm-d, NVIDIA Dynamo, speculative decoding	Days 11, 75
FP8 / FP4 quantisation formats	Day 8
OpenTelemetry GenAI semantic conventions (Development status)	Days 22, 47, 74
Eval-driven development and trajectory evals	Days 12, 35, 44, 49
EU AI Act after the Digital Omnibus (Reg. (EU) 2026/1744)	Day 69
Coding agents as the primary developer interface	Day 80
Managed agent platforms on AWS, Azure and Google Cloud	Day 71

Sources checked (11 Oct 2026)

1. MCP specification 2026-07-28 changelog — `modelcontextprotocol.io/specification/2026-07-28/changelog`
2. Agentic AI Foundation formation (Linux Foundation, 9 Dec 2025) — `linuxfoundation.org/press/linux-foundation-announces-the-formation-of-the-agentic-ai-foundation`
3. A2A protocol one-year milestone (Linux Foundation, 9 Apr 2026) — `linuxfoundation.org/press/a2a-protocol-surpasses-150-organizations-lands-in-major-cloud-platforms-and-sees-enterprise-production-use-in-first-year`
4. OWASP Top 10 for Agentic Applications 2026 — `genai.owasp.org/resource/owasp-top-10-for-agentic-applications-for-2026`
5. Agent Skills open standard — `agentskills.io`
6. EU Digital Omnibus on AI enters into force (White & Case) — `whitecase.com/insight-alert/eu-ai-omnibus-enters-force-amending-ai-act`
7. OpenTelemetry GenAI semantic conventions status (Aug 2026) — `truefoundry.com/blog/opentelemetry-genai-semantic-conventions`
8. llm-d prefill/decode disaggregation — `llm-d.ai/docs/guides/pd-disaggregation`
9. Harness engineering for coding agent users (B. Bockeler, Apr 2026) — `martinfowler.com`

1. Merge Evidence — Where the Incoming Syllabus Overlaps Weeks 1–4

Before placing anything, the incoming syllabus was measured term-by-term against the Weeks 1–4 lesson content (measured 2 August 2026).

Incoming block	Covered	Total	%	Verdict
M2W3 Transformers & LLMs	7	8	88%	COVERED IN WEEKS 1–4
M1W2 SQL	6	7	86%	COVERED IN WEEKS 1–4
M3W3 RAG + Agentic Frameworks	12	16	75%	COVERED IN WEEKS 1–4
M4W3 Cloud Infrastructure	9	12	75%	COVERED IN WEEKS 1–4
M3W4 Production AI & Evaluation	8	11	73%	COVERED IN WEEKS 1–4
M3W2 LLM & Prompt Engineering	10	14	71%	COVERED IN WEEKS 1–4
M3W1 Software Engineering / FastAPI	7	12	58%	PARTIAL
M2W3 Classical NLP	6	12	50%	PARTIAL
M1W3 ML Fundamentals	7	16	44%	PARTIAL
M2W1 Deep Learning / CNN	5	13	38%	PARTIAL
M1W1 Python / Pandas / NumPy / OOP	2	8	25%	**NEW**
M4W4 No-Code & AI Productivity Tools	2	14	14%	**NEW**

Conclusion: roughly **60% of the incoming syllabus is covered by Weeks 1–4**. The genuine additions are classical ML, deep learning and computer vision, classical NLP, software-engineering/FastAPI depth, named agentic frameworks, and the entire no-code / AI-productivity block.

2. Placement Strategy — Architect's Rationale

Three moves make a 16-week syllabus fit 12 weeks without deleting anything:

Move 1 — Deduplicate. Topics taught to production depth in Days 1–28 are not repeated in Weeks 5–12. They are cross-referenced to the day that teaches them, so nothing is lost and no time is spent twice.

Move 2 — Prerequisites become a self-study Foundations Track. Python, Pandas, NumPy, OOP and SQL are upstream of Day 1 — the programme assumes working Python and SQL from the first lesson. They are preserved in full as **Track 0** with all resources, as reference and revision material, not as teaching days.

Move 3 — Keep Weeks 5–12 lean where Weeks 1–4 already go deep. Security had a full production day on Day 25 (Tier 1–4 control hierarchy, injection, PII, tenant isolation, audit, supply chain), so Week 10 keeps red teaming and governance but does not repeat the foundations. The same logic applies to multimodal (Day 26) and agentic system design (Day 20).

What was compressed, stated plainly

Original plan	Now	Justification
W8 Agentic System Design — 7 days	3 days (D54–56)	Day 20 covered orchestration, pools, parallel phases
W9 Multimodal — 7 days	2 days (D67–68)	Day 26 covered VLM architecture, pipelines, tables, staged eval
W10 Security — 7 days	2 days (D69–70)	Day 25 covered the full Tier 1–4 control hierarchy
W12 Capstone — 7 days	4 days (D81–84)	3 days redirected to the no-code block

This is a real trade. Depth in security, multimodal and system design is reduced to create room for ML/DL/NLP foundations and the no-code block. The compressed material is not removed — it layers on top of Days 20, 25 and 26.

TRACK 0 · FOUNDATIONS (Self-Study — Not Teaching Days)

Preserved verbatim from the incoming syllabus. Reference and revision material. No day numbers assigned; these are upstream of Day 1.

Foundations Week 1 · Python Setup, Core Programming & Pandas Basics

Topics

- Python Installation & Environment Setup
- Explore IDEs: VS Code, Jupyter, Colab
- Jupyter Notebook Basics
- Python Basics: Syntax, Variables, Strings, Input/Output
- Loops, Functions, Conditional Statements
- Python Libraries Exploration: Pandas, NumPy, Seaborn, Plotly
- Python Data Structures: Lists, Tuples, Dictionaries, Sets
- DataFrames, Series, Indexing, Selection
- Data Input/Output
- Data Cleaning Techniques
- Handling Missing Data
- Data Visualization with Matplotlib, Seaborn, Plotly
- NumPy Arrays, Broadcasting, Indexing
- OOPS Basics, Classes, Objects, Constructors, Data Input/Output
- Inheritance, Polymorphism, Encapsulation, Abstraction

Free Resources

1. Python Tutorial For Beginners (13 Hours) *(AI Coach John's Material)*

GenAI Support: ChatGPT

Foundations Week 2 · Structured Query Language

Topics

- Intro to SQL, Basic SQL syntax, data types, primary keys, and constraints
- Creating and Inserting into DB
- DQL and DML Commands and filtering data
- Aggregate Functions and Subqueries
- Group By and Joins, types of joins
- Window Functions and Stored Procedures

Free Resources

1. SQL Full Course (4.5 Hours) *(AI Coach John's Material)*

GenAI Support: ChatGPT

Applied in the programme: SQL is exercised continuously in the Days 13–27 Text-to-SQL capstone — joins, aggregates, subqueries, window functions and CTEs all appear in generated and validated queries.

WEEKS 1–4 · DAYS 1–28 · LLM FOUNDATIONS → PRODUCTION

12 October – 8 November 2026

Day-by-Day

Day	Date	Topic
1	Mon Oct 12	Tokenization
2	Tue Oct 13	Embeddings
3	Wed Oct 14	Transformer Architecture
4	Thu Oct 15	Training Lifecycle
5	Fri Oct 16	Scaling Laws & Prompt Engineering
6	Sat Oct 17	Structured Output
7	Sun Oct 18	**Week 1 Project** — Model Selection Dashboard
8	Mon Oct 19	Quantization
9	Tue Oct 20	LoRA & PEFT
10	Wed Oct 21	Full Fine-Tuning
11	Thu Oct 22	Model Serving
12	Fri Oct 23	Evaluation
13	Sat Oct 24	Efficiency
14	Sun Oct 25	**Week 2 Project** — Text-to-SQL Foundation (running capstone begins)
15	Mon Oct 26	RAG Foundations
16	Tue Oct 27	RAG Indexing Pipelines
17	Wed Oct 28	RAG Retrieval Deep Dive
18	Thu Oct 29	RAG Generation & Prompt Engineering
19	Fri Oct 30	Agents and Tool Use
20	Sat Oct 31	Multi-Agent Systems
21	Sun Nov 1	**Week 3 Capstone** — End-to-End Text-to-SQL
22	Mon Nov 2	Production LLM Operations
23	Tue Nov 3	Prompt and Model Versioning
24	Wed Nov 4	Cost Optimization at Scale
25	Thu Nov 5	Security and Compliance
26	Fri Nov 6	Multi-Modal Systems
27	Sat Nov 7	Fine-Tuning for Production
28	Sun Nov 8	**Week 4 Capstone** — Production Hardening & Review

2026 Currency Layer (u1) — added inside each day, on top of the topic above

Day	2026 addition
1	Tokenizer differences across model families; token-cost tax on non-English text (Tamil vs English)
2	Matryoshka (truncatable) embeddings; multi-vector late interaction (ColBERT); multimodal embeddings
3	Mixture-of-Experts (total vs active parameters); GQA / MLA attention; long-context mechanics
4	The 2026 post-training stack: SFT → preference optimisation (DPO) → RL with verifiable rewards (RLVR / GRPO); how reasoning models are made
5	Test-time compute scaling; reasoning-effort controls; prompting reasoning models vs classic models
6	Constrained decoding and strict JSON-Schema output modes; JSON Schema 2020-12
7	Live 2026 model landscape (frontier vs open-weight, verified from official changelogs on the day); model churn and migration cost; model routing
8	FP8 and 4-bit floating-point formats (NVFP4 / MXFP4); KV-cache quantisation
9	QLoRA; serving many LoRA adapters on one base model (multi-LoRA)
10	RL fine-tuning: RLVR / GRPO, verifier and reward design; open tooling (TRL, verl, OpenRLHF)

Day	2026 addition
11	vLLM, SGLang, TensorRT-LLM; prefill/decode disaggregation; llm-d (CNCF Sandbox, Mar 2026) and NVIDIA Dynamo; speculative decoding; prefix caching
12	Benchmark saturation and contamination; agentic benchmarks (e.g. SWE-bench Verified); building your own task evals
13	Prompt caching, batch APIs, KV-cache reuse, model cascades
14	Project add-on: eval set + cost report per candidate model
15	Decision framework: RAG vs long context vs agentic search
16	Contextual chunk enrichment; late chunking; layout-aware document parsing
17	Multi-vector retrieval; GraphRAG; query rewriting and routing
18	Grounded citations; agentic RAG (the agent decides when and what to retrieve)
19	Tool calling with reasoning models; computer-use and browser agents
20	Orchestrator-worker and sub-agent patterns; preview of the A2A protocol
21	Capstone add-on: tool-call trace review and failure taxonomy
22	OpenTelemetry GenAI semantic conventions (still 'Development' status in 2026); AI gateways
23	Prompt registries; model-deprecation and migration playbook for fast model churn
24	Prompt caching, batch inference, routing and reasoning-effort budgets as cost levers
25	OWASP Top 10 for LLM Applications (2025) + OWASP Top 10 for Agentic Applications 2026 (ASI01-ASI10, published 9 Dec 2025); indirect prompt injection
26	Real-time voice agents (speech-to-speech); video understanding
27	Distillation from frontier models; synthetic data; fine-tune vs context-engineer decision
28	Review add-on: OWASP LLM + Agentic risk mapping of the capstone

Week	Days	Theme
1	1–7	Foundations — tokenization, embeddings, transformers, training, scaling laws, structured output, model selection
2	8–14	Efficiency — quantization, LoRA/PEFT, full fine-tuning, serving, evaluation, efficiency, Text-to-SQL foundation
3	15–21	Retrieval & Agents — RAG foundations, indexing, hybrid retrieval, generation, agents, multi-agent, capstone
4	22–28	Production — LLMOps, versioning, cost, security, multimodal, fine-tuning, Week 4 capstone

Incoming topics covered in Weeks 1–4

Incoming topic	Taught on
Tokenization	Day 1
Word Embeddings; Cosine Similarity, Semantic Search & Text Similarity	Day 2
Attention Mechanism, Self-Attention & Encoder-Decoder Architecture; Transformer Architecture & Modern Language Models	Day 3
Large Language Models (LLMs) & Foundation Models	Days 3–4
LLM Fundamentals & Inference Pipeline	Days 4, 11
Prompt Engineering Principles & Prompt Patterns; Zero-Shot, One-Shot & Few-Shot Prompting; Chain-of-Thought	Day 5
Structured Outputs; Output Parsers & Schema Validation	Day 6
Decoding Parameters (Temperature, Top-P, Max Tokens)	Days 5–6
Function Calling & Tool Calling	Days 6, 19
Retrieval-Augmented Generation (RAG); Document Loaders & Chunking Strategies	Days 15–16
Embedding Models & Vector Databases	Days 15–16
Semantic Search & Reranking	Day 17
Agent Memory & Planning Strategies	Days 19–20
Prompt Injection & Prompt Security	Days 18, 25
AI Safety & Guardrails	Days 22, 25
Building Production-Ready LLM Applications	Days 14, 21, 22
Docker for AI Applications *(introduced)*	Day 11
Context Window & Token Management *(cost framing)*	Day 24

Free Resources for blocks covered in Weeks 1–4

From M3W2 · LLM Engineering & Prompt Engineering

1. ☐35 LPA LLM Engineer Roadmap 2026 in Tamil (Beginner to Job Ready) *(AI Coach John)*
2. Prompt Engineering Basics to Advanced: One-Shot, Two-Shot, Chain of Prompting — Tamil Full Guide *(AI Coach John)*
3. Prompt Injection Simplified in Tamil *(AI Coach John)*

GenAI Support: ChatGPT

From M3W3 · RAG Systems *(retrieval half — taught Days 15–18)*

1. 8 RAG Concepts Explained in 26 Minutes: April 2026 *(AI Coach John)*
2. Vector Database (தமிழில்) — The Why? What? How? *(AI Coach John)*
3. 8 (CHUNKING Strategies) Industries Expect You to Know *(AI Coach John)*

GenAI Support: ChatGPT

The agentic-framework half of M3W3 (LangChain, LangGraph, CrewAI, AG2/AutoGen/Agno, Loop Engineering & Human-in-the-Loop) is taught in Weeks 5–6; its resources are listed there.

WEEKS 5–12 · DAYS 29–84 · AGENTS → FOUNDATIONS → CLOUD → CAPSTONE

9 November 2026 – 3 January 2027

WEEK 5 · AGENTS, HARNESS & LOOP ENGINEERING + AGENTIC FRAMEWORKS

Days 29–35 · 9–15 November 2026
Project: AI Research Agent with Tool Calling (PORTFOLIO)

Day	Date	Topic
29	Mon Nov 9	Agent Architectures Beyond ReAct
30	Tue Nov 10	**Harness Engineering I — Tool Design**
31	Wed Nov 11	**Harness Engineering II — Responses, Errors & Environment**
32	Thu Nov 12	**Loop Engineering I — Termination & Budgets**
33	Fri Nov 13	**Loop Engineering II — Recovery, Sub-Agents & Human-in-the-Loop Design**
34	Sat Nov 14	**Agentic Frameworks I — LangChain Fundamentals & LangGraph Workflows**
35	Sun Nov 15	**Week 5 Project** — AI Research Agent with Tool Calling

2026 Currency Layer (u1) — added inside each day, on top of the topic above

Day	2026 addition
29	'Agent = Model + Harness' (Bockeler, Thoughtworks, Apr 2026); long-horizon / deep agents; planning patterns
30	Agent Skills open standard (SKILL.md, progressive disclosure, agentskills.io); AGENTS.md project instructions
31	Sandboxing and containment for agents with shell, web and file access; least-privilege tool permissions
32	Token, time and cost budgets per run; reasoning-effort as a loop control
33	Approval gates and permission modes; interrupt / resume for human review
34	LangGraph durable execution, checkpointing and interrupts
35	Project add-on: trajectory-level eval of the research agent

Incoming topics absorbed: LangChain Fundamentals · LangGraph Workflows · Loop Engineering & Human-in-the-Loop Design · Agent Memory & Planning Strategies *(D33, extended D41)*

Project (from incoming M4W2): *(Project 2)* **AI Research Agent with Tool Calling** — Category: Agentic AI · Industry: Research Automation · YouTube search: *LangGraph AI Research Agent Project*

Free Resources carried in

1. Loop Engineering (தமிழ்) *(AI Coach John)*
2. Complete LangGraph Tutorial in Tamil (Free & Open Source) *(AI with Akash)*
3. LangChain for Beginners: Your First AI Agent Explained in Tamil *(AI Coach John)*

WEEK 6 · CONTEXT ENGINEERING, MCP & MULTI-AGENT FRAMEWORKS

Days 36–42 · 16–22 November 2026
Project: Multi-Agent Business Analyst System (PORTFOLIO)

Day	Date	Topic
36	Mon Nov 16	**Context Engineering I — The Context Budget & Token Management**
37	Tue Nov 17	**Context Engineering II — Compaction & Progressive Disclosure**
38	Wed Nov 18	MCP (Model Context Protocol) Fundamentals
39	Thu Nov 19	Building a Production MCP Server
40	Fri Nov 20	**Agentic Frameworks II — CrewAI, AG2 / AutoGen / Agno & Multi-Agent Collaboration**
41	Sat Nov 21	Agent Memory Architectures & Planning Strategies
42	Sun Nov 22	**Week 6 Project** — Multi-Agent Business Analyst System

2026 Currency Layer (u1) — added inside each day, on top of the topic above

Day	2026 addition
36	Context engineering framework: Write / Select / Compress / Isolate
37	Compaction, memory files and skills as progressive disclosure in practice
38	MCP spec revision 2026-07-28: stateless core (no initialize handshake, no sessions), server/discover, Multi Round-Trip Requests, Tasks extension, Roots/Sampling/Logging deprecated; MCP governed by the Agentic AI Foundation (Linux Foundation, Dec 2025)
39	MCP authorisation (OAuth, Client ID Metadata Documents), cacheable list results (ttlMs / cacheScope), OpenTelemetry trace propagation, migrating a 2025-11-25 server
40	A2A protocol v1.0 (Linux Foundation; Agent Cards, task lifecycle, JSON-RPC + gRPC); MCP vs A2A; vendor agent SDKs (OpenAI Agents SDK, Claude Agent SDK, Google ADK, Microsoft Agent Framework)
41	Long-term memory stores; memory as a tool; memory poisoning risk (OWASP Agentic)
42	Project add-on: expose one agent over A2A and its tools over MCP

Incoming topics absorbed: Context Window & Token Management · MCP (Model Context Protocol) · CrewAI & Multi-Agent Collaboration · AG2 / AutoGen / Agno Framework Overview · Agent Memory & Planning Strategies

Project (from incoming M4W2): *(Project 3)* **Multi-Agent Business Analyst System** — Category: Multi-Agent AI · Industry: Business Intelligence · YouTube search: *CrewAI Multi Agent Project*

Free Resources carried in

1. MCP Crash Course (2 Hours) *(AI Coach John)*
2. Build AI Agents with Crew AI Framework *(AI with Thiru)*
3. Autogen Agentic AI Framework *(Aai Tech)*

WEEK 7 · PRODUCTION AI SYSTEMS, EVALUATION & OBSERVABILITY

Days 43–49 · 23–29 November 2026
Project: Eval & Observability Platform (PORTFOLIO)

Day	Date	Topic
43	Mon Nov 23	**AI System Architecture & Orchestration**
44	Tue Nov 24	LLM-as-a-Judge & Eval Harness Design
45	Wed Nov 25	**Hallucination Detection & Evaluation**
46	Thu Nov 26	**RAG Evaluation — RAGAS & TruLens**
47	Fri Nov 27	**LangSmith for Debugging & Observability**
48	Sat Nov 28	**AI Safety & Guardrails in Production**
49	Sun Nov 29	**Week 7 Project** — Eval & Observability Platform

2026 Currency Layer (u1) — added inside each day, on top of the topic above

Day	2026 addition
43	Reference architecture: gateway → guardrails → orchestrator → tools → observability
44	Eval-driven development; golden datasets; agent trajectory and tool-use evals
45	Groundedness and citation checks; uncertainty signals
46	Agentic RAG evaluation (retrieval decisions, not just answers)
47	Vendor-neutral tracing with OpenTelemetry GenAI conventions alongside LangSmith (Langfuse, Arize Phoenix)
48	Guardrails mapped to OWASP Agentic risks; runtime policy enforcement
49	Project add-on: CI eval gate that blocks a regression

Incoming topics absorbed: AI System Architecture & Orchestration · LangSmith for Debugging & Observability · AI Safety & Guardrails · Hallucination Detection & Evaluation · RAG Evaluation (RAGAS, TruLens)

WEEK 8 · AI APPLICATION ENGINEERING & AGENTIC SYSTEM DESIGN

Days 50–56 · 30 November – 6 December 2026
Project: End-to-End AI Backend + Scalable Agent Runtime (PORTFOLIO)

Day	Date	Topic
50	Mon Nov 30	**AI Application Architecture · REST APIs, HTTP Methods & API Integration · JSON, JSON Schema & Structured Data**
51	Tue Dec 1	**FastAPI Fundamentals & API Development · Request Handling, Response Models & Error Handling**
52	Wed Dec 2	**Authentication, API Keys & Environment Variables · File Uploads, Streaming Responses & Async APIs**
53	Thu Dec 3	**Database Integration (SQLite/PostgreSQL Basics) · AI API Integration (OpenAI, Gemini, Claude)**
54	Fri Dec 4	Agentic System Design — Runtime, State Management & Queueing
55	Sat Dec 5	Scaling, Backpressure, Failure Isolation & Concurrency
56	Sun Dec 6	**Week 8 Project** — End-to-End AI Backend Development using FastAPI + Scalable Agent Runtime

2026 Currency Layer (u1) — added inside each day, on top of the topic above

Day	2026 addition
50	Streaming tool-call events; structured outputs at the API boundary
51	Async FastAPI patterns for long-running LLM calls
52	Server-Sent Events streaming; secrets management for model API keys
53	Multi-provider abstraction and fallbacks; provider-native MCP connectors
54	Durable execution for long-running agents (workflow engines such as Temporal); resumable state
55	Rate limits, retries with jitter, circuit breakers for model providers
56	Project add-on: provider failover drill

Incoming topics absorbed — the complete M3W1 block: Introduction to AI Application Architecture · REST APIs, HTTP Methods & API Integration · JSON, JSON Schema & Structured Data · FastAPI Fundamentals & API Development · Authentication, API Keys & Environment Variables · Request Handling, Response Models & Error Handling · File Uploads, Streaming Responses & Async APIs · Database Integration (SQLite/PostgreSQL Basics) · AI API Integration (OpenAI, Gemini, Claude) · End-to-End AI Backend Development using FastAPI

Free Resources carried in

1. API — Step by Step — Build, Secure and Deploy | Full Practical Guide *(Applied with AI)*
2. Complete FastAPI Tutorial in Tamil — Build your own RestAPI *(AI with Akash)*

WEEK 9 · MACHINE LEARNING & DEEP LEARNING FOUNDATIONS

Days 57–63 · 7–13 December 2026
Project: ML & DL Project Development & Deployment (PORTFOLIO)

Day	Date	Topic
57	Mon Dec 7	**ML Fundamentals — Types of ML (Supervised, Unsupervised & Reinforcement) · Features, Labels, Training/Testing Data & Train/Test Split · Classification, Regression & Clustering Algorithms**
58	Tue Dec 8	**Model Validation using Cross-Validation · Evaluation Metrics: Accuracy, Precision, Recall, F1-Score & ROC-AUC · Confusion Matrix & Handling Class Imbalance**
59	Wed Dec 9	**Overfitting, Underfitting & Regularization · Feature Engineering & Data Preprocessing · Feature Scaling, Normalization & Data Preparation · End-to-End ML Workflow using Scikit-learn**
60	Thu Dec 10	**Deep Learning Fundamentals — Neural Networks & Perceptron · Activation Functions, Loss Functions & Optimizers · Forward Propagation, Backward Propagation & Gradient Descent**
61	Fri Dec 11	**Epochs, Batch Size, Learning Rate & Hyperparameter Tuning · Batch Normalization, Dropout & Vanishing Gradient Problem**
62	Sat Dec 12	**CNNs & Image Classification · Pretrained Models, Transfer Learning & Fine-Tuning · SOTA Models & Modern DL Applications · End-to-End Model Development using TensorFlow/Keras or PyTorch**
63	Sun Dec 13	**Week 9 Project** — ML & DL Project Development & Deployment

2026 Currency Layer (u1) — added inside each day, on top of the topic above

Day	2026 addition
57	Where classical ML still runs FAANG-scale systems (ranking, recommendations, fraud)
58	Metric choice for imbalanced production data
59	Data leakage and train/serve skew
60	From perceptron to transformer: the same gradient-descent engine
61	Mixed-precision training basics (FP16 / BF16)
62	Vision Transformers (ViT) and diffusion models in context
63	Project add-on: model card for each project

Incoming topics absorbed — the complete M1W3 and M2W1 blocks.

Week 9 Project Set — preserved verbatim

ML Project Development & Deployment *(incoming M1W4)*

Project Title	Category	Industry / Application
Predictive Maintenance for Industrial Equipment	Regression	Manufacturing / Industry 4.0
Credit Risk Assessment & Loan Default Prediction	Classification	Banking & FinTech
Customer Segmentation using K-Means & DBSCAN	Clustering	Retail / E-Commerce / Marketing

DL Project Development & Deployment (Image/Video) *(incoming M2W2)*

Project Title	Category	Industry / Application
Brain Tumor Detection & Segmentation using U-Net + CNN	Image Classification / Segmentation	Healthcare / Medical AI
Video Anomaly Detection for Smart Surveillance	Video Classification / Anomaly Detection	Smart Cities / Security

Free Resources carried in

1. Machine Learning (தமிழில்) in 25 Minutes: Absolute Beginner Crash Course *(AI Coach John's Material)*
2. Industry-Standard AI, Data Science & Machine Learning Full Course in Tamil for Beginners *(Hire Ready)*
3. AI/ML (தமிழில்): Complete Deep Learning RoadMap Explained in 15 Mins *(AI Coach John's Material)*
4. Deep Learning Series in Tamil *(Adi Explains)*

WEEK 10 · NLP, MULTIMODAL, REASONING & RED TEAMING

Days 64–70 · 14–20 December 2026
Project: NLP & Multimodal Projects + Red Team Assessment (PORTFOLIO)

Day	Date	Topic
64	Mon Dec 14	**NLP Foundations — Introduction to NLP & Text Preprocessing · Tokenization, Stop Words, Stemming & Lemmatization**
65	Tue Dec 15	**Bag of Words (BoW), Count Vectorizer & TF-IDF · N-Grams, POS Tagging & Named Entity Recognition (NER)**
66	Wed Dec 16	**Word Embeddings: Word2Vec, GloVe & Vector Embeddings · End-to-End NLP Pipeline using Hugging Face Transformers**
67	Thu Dec 17	Vision-Language Models, OCR Pipelines & Document Understanding at Scale
68	Fri Dec 18	Chart & Diagram Reasoning · Reasoning Models & Test-Time Compute
69	Sat Dec 19	Red Teaming, OWASP LLM Top 10 & Responsible AI Governance
70	Sun Dec 20	**Week 10 Project** — NLP & Multimodal Projects + Red Team Assessment

2026 Currency Layer (u1) — added inside each day, on top of the topic above

Day	2026 addition
64	Why classical preprocessing still matters before LLM pipelines (cost, PII scrubbing)
65	Classical NER vs LLM extraction: cost and accuracy trade-off
66	Hugging Face Hub model selection; small open models for on-device and edge
67	Document AI with vision-language models replacing OCR-only pipelines
68	Reasoning models in production: effort settings, latency/cost trade-offs, when not to use them
69	EU AI Act after the Digital Omnibus (Reg. (EU) 2026/1744, in force 27 Jul 2026): Annex III high-risk → 2 Dec 2027, Annex I → 2 Aug 2028, Art. 50 transparency still from Aug 2026; NIST AI RMF; ISO/IEC 42001; OWASP Agentic red-team cases
70	Project add-on: red-team agent tool misuse, not only prompt injection

Incoming topics absorbed — the complete M2W3 classical-NLP block. *(Attention, self-attention, encoder-decoder, transformers, LLMs, foundation models, cosine similarity and semantic search from the same block are taught on Days 2–4.)*

Week 10 Project Set — preserved verbatim

NLP Projects & Deployment *(incoming M2W4)*

Project Title	Category	Industry / Application
AI Resume Screening & Candidate Matching System	NLP + Semantic Search	HRTech / Recruitment
Legal Document Question Answering using RAG	LLM + RAG	LegalTech / Enterprise AI
Intelligent Customer Support Chatbot with Vector Search	Conversational AI + LLM	Customer Support / SaaS

Also this week: Multimodal Document Processor · Red Team Your Own RAG System (20+ adversarial prompts, guardrails blocking ≥90%, written security assessment with risk ratings and residual-risk acceptance).

Free Resources carried in

1. What is NLP? Beginners to Advanced RoadMap 2025 in Tamil *(AI Coach John's Material)*
2. NLP Playlist *(Chill and Grow)*

WEEK 11 · CLOUD DEPLOYMENT, AI INFRASTRUCTURE & LLMOPS

Days 71–77 · 21–27 December 2026
Project: Deploy a Production-Ready AI Service (PORTFOLIO)

Day	Date	Topic
71	Mon Dec 21	**Introduction to AWS, Azure & Google Cloud Platform · Compute Services (EC2, Azure VM, Compute Engine) · Storage Services (S3, Blob Storage, Cloud Storage)**
72	Tue Dec 22	**Docker Fundamentals & Containerization · Docker for AI Applications**
73	Wed Dec 23	**Deploying FastAPI Applications · Deploying AI Applications on Cloud · Serverless AI Deployments**
74	Thu Dec 24	**CI/CD Basics for AI Applications · Monitoring & Logging · Cost Optimization & Cloud Best Practices**
75	Fri Dec 25	LLMOps — Model Registries, Automated Eval Gates & Inference Optimisation (TensorRT-LLM, ONNX)
76	Sat Dec 26	Synthetic Data Generation (Self-Instruct, Evol-Instruct) · Feature Stores & Data Lineage · Cost Engineering & Chargeback
77	Sun Dec 27	**Week 11 Project** — Deploy a Production-Ready AI Service

2026 Currency Layer (u1) — added inside each day, on top of the topic above

Day	2026 addition
71	Managed agent platforms: Amazon Bedrock AgentCore, Microsoft Azure AI Foundry, Google Vertex AI agent tooling
72	GPU containers and model-weight caching
73	Serverless GPU inference; cold-start trade-offs
74	Eval gates inside CI/CD; OpenTelemetry-based AI monitoring
75	Kubernetes-native inference (llm-d, NVIDIA Dynamo) and prefill/decode disaggregation at scale
76	Synthetic data for evals and RL environments; verifier quality as the ceiling
77	Project add-on: cost-per-request dashboard

Incoming topics absorbed — the complete M4W3 block plus M3W4's Docker and Cloud Deployment.

Free Resources carried in

1. AWS in Half Hour | Learn EC2 Basics in Tamil *(Tamil Coding / AWS Tamil)*
2. What is Docker? Explained in Tamil | Full Tutorial + Demo *(Arivi by HCL GUVI)*
3. AWS EC2 vs Azure Virtual Machine (தமிழில்) *(K. Mohamed Faizal)*
4. Docker & Containerization in Tamil *(Arivi by HCL GUVI)*
5. AWS for Beginners in Tamil Playlist *(Tamil Coding)*
6. Azure Fundamentals in Tamil *(K. Mohamed Faizal)*
7. Google Cloud Platform (GCP) in Tamil *(Tamil Cloud / Community Resources)*
8. CI/CD Pipeline in Tamil *(DevOps Tamil)*
9. FastAPI Deployment using Docker & Cloud *(English — Recommended if Tamil is unavailable)*
10. Deploying AI Applications on AWS using FastAPI & Docker *(English — Advanced)*
11. Dockers for Beginners *(Simply Byte)*

WEEK 12 · NO-CODE AI PLATFORMS & ENTERPRISE CAPSTONE

Days 78–84 · 28 December 2026 – 3 January 2027
Project: Enterprise AI Assistant (CAPSTONE)

Day	Date	Topic
78	Mon Dec 28	**Introduction to No-Code & Low-Code AI Platforms · Claude Projects & Claude Artifacts · ChatGPT Projects, GPTs & Canvas**
79	Tue Dec 29	**Microsoft Copilot & Microsoft 365 Copilot · Microsoft Copilot Studio & AI Agents (Basics) · Google Gemini Workspace & Gems**
80	Wed Dec 30	**AI Workflow Automation using n8n (Introduction) · AI Coding Assistants (GitHub Copilot, Cursor & Windsurf) · Rapid AI Prototyping & Business Use Cases · Choosing Between No-Code, Low-Code & Custom AI Development**
81	Thu Dec 31	**Capstone I** — Architecture, ADRs & C4 Diagrams
82	Fri Jan 1	**Capstone II** — Build & Integration
83	Sat Jan 2	**Capstone III** — Evaluation, Security & Deployment
84	Sun Jan 3	**Capstone IV** — Enterprise AI Assistant Final Defence

2026 Currency Layer (u1) — added inside each day, on top of the topic above

Day	2026 addition
78	Agent builders inside productivity suites; when no-code agents are enough
79	Enterprise agent governance in Microsoft 365 / Google Workspace
80	Coding agents as the main developer interface (Claude Code, OpenAI Codex, GitHub Copilot agent mode, Cursor); Agent Skills and AGENTS.md in a real repo
81	ADRs for MCP, A2A, model routing and eval-gate choices
82	Build with MCP tools, Agent Skills and OpenTelemetry tracing
83	Eval gate + OWASP LLM/Agentic mapping + EU AI Act risk classification
84	Final defence + State of AI Engineering 2026 → 2027 outlook

Incoming topics absorbed — the complete M4W4 block.

Capstone Project — incoming Enterprise Project 1

Project Title	Category	Industry / Application	YouTube Search
(Project 1) **Enterprise Multi-Document RAG Assistant**	RAG + LangChain	Enterprise Knowledge Management	Enterprise RAG LangChain LangGraph Project

Capstone Required Deliverables

1. **Architecture Decision Records (ADRs)** for every major choice (vector DB, model, framework, cloud)
2. **C4 System Diagrams** — Context, Container, Component, Code
3. **Non-Functional Requirements** — latency SLAs, availability, cost/request, scaling limits
4. **Risk Assessment** — failure modes, guardrail-bypass scenarios, residual risk
5. **Deployment Plan** — blue-green strategy, rollback, monitoring dashboards
6. **Security Report** — red-team results, guardrail coverage, compliance mapping

Free Resources carried in

1. Claude Projects & Claude Artifacts (தமிழில்) *(AI Coach John)*
2. Google Gemini Workspace & Gems (தமிழில்) *(AI Coach John)*
3. Microsoft Copilot & Microsoft 365 Copilot (தமிழில்) *(K. Mohamed Faizal)*
4. Microsoft Copilot Studio & AI Agents (Basics) *(English — Microsoft Official)*
5. AI Workflow Automation using n8n (தமிழில்)
6. AI Workflow Automation using n8n (தமிழில்) — Part 2
7. GitHub Copilot (தமிழில்) *(Tamil Tech / Tamil Coding Community)*

3. Complete Placement Index — Every Incoming Block

Incoming	Title	Placed at
M1W1	Python Setup, Core Programming & Pandas Basics	**Track 0** (self-study)
M1W2	Structured Query Language	**Track 0** (self-study); applied Days 13–27
M1W3	Machine Learning Fundamentals & Core Concepts	**Days 57–59**
M1W4	ML Project Development & Deployment	**Day 63**
M2W1	Deep Learning Fundamentals & Neural Networks	**Days 60–62**
M2W2	DL Project Development & Deployment (Image/Video)	**Day 63**
M2W3	NLP & Large Language Models	**Days 64–66** (classical) + Days 1–4 (transformers/LLMs)
M2W4	NLP Projects & Deployment	**Day 70**
M3W1	Software Engineering Foundations for AI Applications	**Days 50–53**
M3W2	LLM Engineering & Prompt Engineering	Days 4–6, 18, 25 + **Day 36** (token mgmt)
M3W3	RAG Systems & Agentic AI Frameworks	Days 15–20 + **Days 33, 34, 40, 41** (frameworks, loop, HITL)
M3W4	Production AI Systems, Evaluation & Deployment	**Days 43–48**, 72, 73
M4W1	Enterprise AI Project Development — 1	**Week 12 Capstone**
M4W2	Enterprise AI Project Development — 2	**Days 35, 42**
M4W3	Cloud Deployment & AI Infrastructure	**Days 71–74**
M4W4	No-Code AI Platforms & AI Productivity Tools	**Days 78–80**

Every incoming block is placed. Nothing dropped. No weeks added.

4. Project Ladder

Week	Project	Type
01	Model Selection Dashboard	MINI
02	Domain Fine-Tune + Quantize	PORTFOLIO
03	End-to-End Text-to-SQL	**CAPSTONE**
04	Production Hardening & Review	PORTFOLIO
05	AI Research Agent with Tool Calling	PORTFOLIO
06	Multi-Agent Business Analyst System	PORTFOLIO
07	Eval & Observability Platform	PORTFOLIO
08	End-to-End AI Backend + Scalable Agent Runtime	PORTFOLIO
09	ML & DL Project Development (5 projects)	PORTFOLIO
10	NLP & Multimodal Projects + Red Team Assessment (3 projects)	PORTFOLIO
11	Deploy a Production-Ready AI Service	PORTFOLIO
12	Enterprise Multi-Document RAG Assistant	**CAPSTONE**

13 named projects across the programme.

5. Daily Protocol

1. **Teach** the day's topic fully in chat — concept-first, plain English, FAANG examples, Architect's Lens, hands-on exercises
2. **Eagle Eye** — two-pass verification; never declare a gap without snip verification, never declare coverage without confirmation
3. **PDF** → `~/Desktop/AI-ML/WeekN_DayNN_TopicName.pdf`
4. **Push** → `week-NN/notes/` on the GitHub repository

FAANG rule (locked): examples use only Meta, Apple, Amazon, Netflix, Google, Microsoft.
GenAI Support: ChatGPT *(as noted throughout the incoming syllabus)*

Locked 11 October 2026 — fresh run, Day 1 = Mon 12 Oct 2026, Day 84 = Sun 3 Jan 2027. Content merge revision m1 (2 Aug 2026) + 2026 Currency Layer u1 (11 Oct 2026). Complete content authority for Weeks 1–12. Dates governed by `SCHEDULE.md`.

History

- **6 Jul – 2 Aug 2026:** a first run covered Days 1–28, then paused. Its lesson PDFs were removed on 11 Oct 2026 for a clean restart; they remain recoverable from git history (last present at commit `7f258ab`).
- **2 Aug 2026:** content merge revision m1 — a 4-month external syllabus absorbed into the 12 weeks (133/133 topics, 53/53 resources preserved).
- **11 Oct 2026:** fresh run locked — Day 1 = 12 Oct 2026, Day 84 = 3 Jan 2027.
- **11 Oct 2026:** 2026 Currency Layer u1 — one verified 2026 addition per day (MCP 2026-07-28, A2A v1.0, Agent Skills, OWASP Agentic Top 10, RLVR, disaggregated serving, EU AI Act Omnibus). Additive only; no m1 topic changed.

Locked 11 October 2026.